

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a48b9c26cd59ee5d3.jsonl`  
- SHA-256: `c9dea2950f32603c9ce24de90b2aa5bbfb9c4f93eade5f5800a7135d769d337a5`  
- Source modified: `2026-06-16T03:25:53+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:56.525Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer is a single-stage, anchor-free temporal action localization model that combines a multiscale feature pyramid with local self-attention (a Transformer) and a lightweight decoder that classifies every moment in time and regresses the corresponding action boundaries, eliminating the need for separate action proposals."

****Source:**** https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)

****Supporting quote:**** "The model employs "a multiscale feature representation with local self-attention" and "a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries." It operates as a single-stage, anchor-free detector without requiring action proposals."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:02.855Z)

I'll verify this claim about ActionFormer's architecture. Let me check the primary source and look for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:25:07.013Z)

I'll verify this claim against the primary source (the ActionFormer ECCV 2022 paper) and search for any contradicting evidence.

4. user (2026-06-16T03:25:12.126Z)

ActionFormer Architecture

****Single-stage & Anchor-free:**** Yes. The paper describes it as "a simple yet powerful model to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows."

****Multiscale features with local self-attention:**** Yes. ActionFormer "combines a multiscale feature representation with local self-attention."

****Lightweight decoder for temporal classification and boundary regression:**** Yes. The model "uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries."

****Action proposals:**** Eliminated. The abstract explicitly states the approach works "without using action proposals," representing a departure from proposal-based temporal localization methods.

5. user (2026-06-16T03:25:13.577Z)

Web search results for query: "ActionFormer single-stage anchor-free temporal action localization multiscale Transformer local self-attention"

Links: [{"title": "ActionFormer: Transformer-based Action Localization", "url": "https://www.emergentmind.com/papers/2202.07925"}, {"title": "[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/abs/2202.07925"}, {"title": "[2202.07925v1] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/abs/2202.07925v1"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "ActionFormer : Localizing Moments of Actions with Transformers", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"}, {"title": "[PDF] ActionFormer: Localizing Moments of Actions with Transformers | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/ActionFormer:-Localizing-Moments-of-Actions-with-Zhang-Wu/58a3fedc03ab9f5908c077c115eff4c8d2d87660"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/364819471_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/pdf/2512.01298"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}]

Based on the search results, here's what I found about ActionFormer:

Overview

ActionFormer is a simple yet powerful model to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows. The paper introduces a Transformer-based single-stage, anchor-free model that simplifies temporal action localization, reducing computational overhead.

Key Architecture Components

ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries. This strategic adoption of local self-attention enhances long-range temporal dependencies essential for distinguishing action boundaries.

Performance Results

ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points. ActionFormer demonstrates strong results on ActivityNet 1.3 (36.6% average mAP) and EPIC-Kitchens 100 (+13.5% average mAP over prior works).

The model represents a significant advancement in temporal action localization by effectively combining Transformer-based architecture with multiscale feature representations and local self-attention mechanisms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:17.109Z)

Both the primary arXiv source and the search results corroborate every component of the claim. Let me do one adversarial pass against the official ECCV/ECVA PDF to ensure there's no qualification I'm missing (e.g., whether "single-stage/anchor-free" is the authors' framing vs. marketing).

7. user (2026-06-16T03:25:26.501Z)

I appreciate your request, but I cannot extract and quote from this content. The provided text appears to be a PDF file encoded in a compressed/binary format (FlateDecode streams), not readable plaintext. The content shown consists primarily of PDF object definitions and encoded streams rather than human-readable text that I could accurately quote from.

To properly answer your questions about ActionFormer's architecture, I would need:

- A plaintext version of the abstract and introduction
- Or a properly rendered PDF I could access
- Or the original paper in a readable format

If you can provide the text in a readable format, I'd be happy to locate exact quotes about ActionFormer being single-stage, anchor-free, using multiscale features with local self-attention, lightweight decoder properties, and any stated limitations.

[Binary content (application/pdf, 1MB) also saved to C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580322199-5hp0ta.pdf]

8. assistant (2026-06-16T03:25:28.800Z)

The PDF saved locally. Let me read it directly to get exact quotes from the authoritative ECVA version.

9. user (2026-06-16T03:25:29.661Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

10. assistant (2026-06-16T03:25:32.990Z)

The PDF can't be rendered locally without poppler. I already have strong corroboration from the arXiv abstract (the primary peer-reviewed source) and the search aggregation. Let me confirm the arXiv abstract text once more verbatim to nail the exact wording.

11. user (2026-06-16T03:25:37.127Z)

ActionFormer Abstract - Key Concepts

Here is the verbatim quote from the abstract containing the technical details you requested:

"a simple yet powerful model to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows. ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries."

This passage directly addresses all your specified concepts: single-shot detection (no proposals or anchor windows), multiscale features, local self-attention, lightweight decoder, moment classification, and boundary estimation.

12. assistant (2026-06-16T03:25:46.245Z)

The claim is fully verified against the primary source. Every component of the claim maps directly to the verbatim abstract:

- "single-stage" / "single shot" ? supported ("identify actions in time... in a single shot")
- "anchor-free" ? supported ("without... relying on pre-defined anchor windows")
- "eliminating the need for separate action proposals" ? supported ("without using action proposals")
- "multiscale feature pyramid with local self-attention (a Transformer)" ? supported ("combines a multiscale feature representation with local self-attention"; it is a Transformer-based model, ECCV 2022)
- "lightweight decoder that classifies every moment in time and regresses the corresponding action boundaries" ? supported verbatim ("uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries")

The source quality is excellent: peer-reviewed ECCV 2022 paper (Zhang, Wu, Li), primary arXiv source, corroborated across arXiv v1, the ECVA proceedings, and ACM DL. The claim is current (2022, the canonical reference architecture in this subfield), widely cited, and not contested by any credible source. No marketing/cherry-picking concerns ? these are factual architecture descriptions, not performance claims. One minor note: the claim's phrase "multiscale feature pyramid" is a faithful paraphrase of the paper's "multiscale feature representation" (the model does build a feature pyramid), not an overreach.

13. user (2026-06-16T03:25:51.409Z)

Structured output provided successfully

14. assistant (2026-06-16T03:25:53.921Z)

Verification complete. The claim is ****not refuted**** ? it is fully and accurately supported by the primary source (ActionFormer, ECCV 2022, arXiv:2202.07925), with the architecture description matching the paper's abstract verbatim across all five components. Confidence: high.